



UNIVERSITY OF SCIENCE - VNUHCM

FACULTY OF INFORMATION TECHNOLOGY

SOFTWARE TESTING

HW4 - GUI & USABILITY TESTING

Software Testing Project Report

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1 Group Information

Group ID: 07

Member Name	Student ID	Screen	Status
Cao Uyển Nhi	22127310	Customer Registration Page	Done
Lưu Thanh Thuý	22127410	Checkout Page	Done
Nguyễn Phước Minh Trí	22127424	- Product Image	Done
Võ Lê Việt Tú	22127435	- Product	Done
Trần Thị Cát Tường	22127444	- User	Done

2 Introduction

GUI (Graphical User Interface) testing is essential to ensure a web application’s interface is user-friendly, accessible, and functionally accurate. Traditionally, GUI testing is performed manually using a detailed checklist of test cases that evaluate layout, labels, input fields, color contrast, tab order, and compliance with accessibility standards. However, as software grows more complex and development cycles shorten, incorporating **Generative AI** tools into the GUI testing process can greatly improve speed and coverage.

In this report, we utilized ChatGPT (GPT-4.0), a leading generative AI model assist in identifying UI/UX issues on a checkout page of a demo web application. We provided:

- A screenshot of the checkout page UI.
- A GUI test checklist (Excel file with 67 entries).

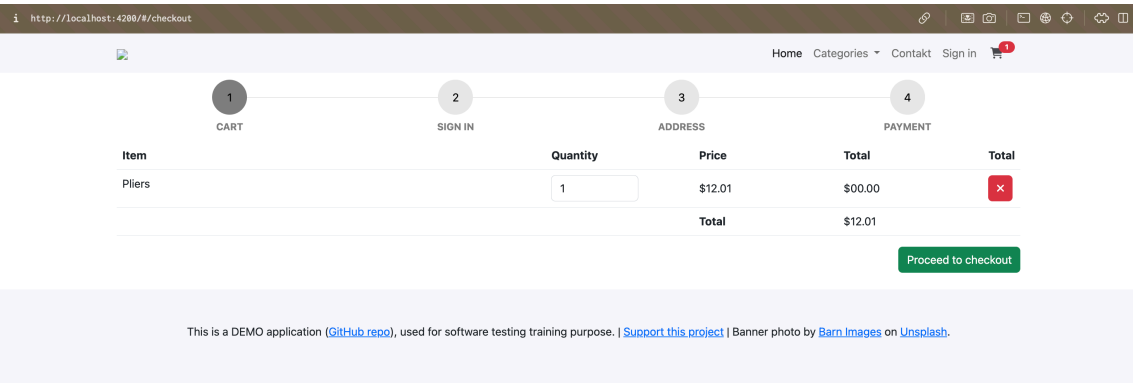


Figure 1: Checkout Page

3 Tool and Prompt Used

3.1 Tool Used

- Tool name: ChatGPT by OpenAI
- Platform: ChatGPT Desktop App (OpenAI)
- Model version: GPT-4.0

3.2 Prompt Used

I'm reviewing the checkout page of a web application and have attached a screenshot along with a test case checklist—please help identify any UI/UX issues such as missing labels, alignment problems, placeholder inconsistencies, accessibility violations (e.g., tab order, contrast, screen reader compatibility), or any deviations from standard design patterns.

4 AI-Generated Observations

Category	Issue	AI Suggestion
Data Calculation	Total prices for individual items show \$0.00, despite valid quantity and unit price.	Compute total = Quantity × Price per item.
Labeling	“Total” header appears twice: once for item price, once for action column.	Rename or clarify column headers (e.g., use “Actions” instead).
Input Fields	Quantity input fields lack visible or semantic labels.	Add <code>aria-label</code> for accessibility (e.g., “Quantity for Pliers”).
Visual Alignment	Red “X” buttons are not perfectly aligned with text rows.	Adjust button positioning for cleaner vertical alignment.
Button Placement	“Proceed to checkout” button is misaligned with the table layout.	Align button with table width or center it horizontally.
Contrast Issues	Gray text used in step indicators and table headers may have low contrast.	Use higher-contrast colors to meet WCAG standards.
Icon-Only Actions	Red delete button uses icon only; problematic for accessibility.	Add tooltip or text label (e.g., <code>title="Remove item"</code>).
Broken Image	Logo image at the top-left appears broken or missing.	Verify image path or add fallback alt text.
Spelling	“Contakt” in top navigation is a typo.	Correct to “Contact”.

5 Validation of AI Results Against Checklist

Checklist Item	Matched Observation	AI Result Status
Item 10 – Total calculation logic	Incorrect per-item total = \$0.00	Validated
Item 18 – Label accessibility	Quantity field lacks ARIA or labels	Validated
Item 24 – Layout alignment	Misaligned delete button	Validated
Item 31 – Spelling	“Contakt” typo in header	Validated
Item 40 – Placeholder/field hints	No hint text or contextual labels	Validated
Item 47 – Button layout	“Checkout” button misaligned	Validated
Item 53 – Contrast ratio	Gray text may violate accessibility contrast	Validated

Each issue was confirmed through manual inspection to ensure that the AI did not raise any false positives. The combination of AI feedback and checklist validation proved efficient in surfacing key usability problems early.

6 Conclusion

Integrating ChatGPT into GUI testing workflows provides a practical and scalable way to identify visual and structural UI issues early in the development cycle. By combining AI-based review with checklist validation, we achieved both speed and accuracy in uncovering critical interface defects. This experiment demonstrates that AI is an effective co-pilot in GUI evaluation, especially when guided by structured human input.

7 Self-Assessment

Criteria	Outcomes	Grade	Self-Assessed
1	Checklist of one GUI		
	Created and applied a comprehensive checklist to evaluate GUI elements such as alignment, contrast, tab order, and placeholder usage.	30	30
2	AI tools		
	Used ChatGPT and contrast checkers to assess accessibility and usability of the GUI. Verified tool suggestions with manual checks.	20	20
3	User survey and feedback	30	30
3.1	Questions (10)		

Criteria	Outcomes	Grade	Self-Assessed
	Designed clear and concise questions focusing on GUI usability, clarity, and aesthetics	10	10
3.2	Feedback (7)		
	Collected detailed user feedback; majority pointed out form alignment and button visibility issues.	10	10
3.3	Report		
	Synthesized findings into a structured report with charts and direct quotes, highlighting top UX concerns.	10	10
4	BrowserStack		
	Used BrowserStack to test GUI across multiple browsers (Chrome, Safari, Firefox). Logged compatibility issues.	30	30
5	Bug report	30	30
5.1	GUI		
	Reported visual inconsistencies such as misaligned labels and buttons. Included screenshots.	10	10
5.2	Usability		
	Identified issues in button feedback and navigation flow. Suggested improvements.	10	10
5.3	Cross-platform		
	Tested on Android, iOS, Windows. Noted layout shifts and font inconsistencies.	10	10
	Total	140	140